SEMESTER 8

INFORMATION TECHNOLOGY

SEMESTER S8

BIOINFORMATICS

Course Code	**PEITT861**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. Students should be able to grasp the foundational principles of molecular biology to effectively navigate and utilize them in research and analysis.
2. To equip students with a range of computational tools and algorithms to perform pairwise and multiple sequence alignments, and construct phylogenetic trees, enabling the identification of evolutionary relationships and genetic variations.
3. Students will be able to utilize statistical and computational algorithms to predict the secondary structures of proteins and RNA, aiding in the understanding of their functions and interactions in biological systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to Bioinformatics** – scope, elementary tasks in bioinformatics, Molecular biology basics - human genome organization, central dogma, DNA and RNA structure, Genetic codes, Datatypes - Gene expression data, Micro Arrays, NGS, Pathways, molecular interactions, Biological Databases - Types of databases, Sequence databases – GenBank, DDBJ, EMBL, Entrez, Unigene, Protein sequence databases – SwissProt, UniProt, PDB	9
2	**Sequence Alignment** – local and global alignment, similarity vs homology, dot matrices and hash coding, FASTA files, Algorithms for pairwise sequence alignment – Needleman and Wunsch, Scoring Matrices – basic concepts, BLAST and its variants, Multiple Sequence Alignment – Goals, scoring, substitution matrices, PAM and BLOSUM, gap penalties, CLUSTAL and its variants	9

3	Phylogenetic trees, Topologies, Distance matrix based and Character based tree construction, Algorithms for Phylogenetic tree construction – UPGMA, Neighbor-Joining, Maximum Parsimony, Pattern representations – deterministic and probabilistic patterns, algorithms for pattern discovery – HMM, gene discovery using GeneMark	**9**
4	**Introduction to Protein structures** – primary, secondary and tertiary, measures of prediction accuracy, statistical algorithms for protein secondary structure prediction – Chou-Fasman, Protein Folding – Lattice models, algorithms for predicting RNA secondary structure – Nussinov algorithm	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain basic concepts in molecular biology and familiarize various biological datatypes and databases	**K2**
CO2	Discuss various algorithms and tools for performing pairwise and multiple sequence alignment	**K2**
CO3	Apply various algorithms for the construction of phylogenetic trees and familiarize basic algorithms for gene discovery	**K3**
CO4	Apply various statistical algorithms for predicting protein and RNA secondary structure	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2			3	3						
CO2	3	2	3		3	3	2					3
CO3	3	2	3		3	3	2					3
CO4	3	3	3		3	3	2					3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Bioinformatics Databases and Algorithms	N Gautham	Narosa Publications	2006
2	Fundamental Concepts of Bioinformatics	D. E. Krane , M. L. Raymer	Pearson Education	2003
3	Bioinformatics: Basics, Algorithms and Applications	Ruchi Singh, Richa Sharma	Universities Press	2010

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	An Introduction to Bioinformatics Algorithms	Neil C Jones and Pavel A Pevzner	MIT Press	2004
2	Introduction to Bioinformatics	T. K. Attwood and D. J. Parry-Smith	Pearson	2003
3	Bioinformatics: Sequence and Genome Analaysis	David W. Mount	CBS Publishers	2005

Video Links (NPTEL, SWAYAM…)	
Sl. No	**Link ID**
1	https://onlinecourses.nptel.ac.in/noc21_bt06/preview

SEMESTER S7

SOFTWARE TESTING

Course Code	PEITT862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCITT503: Software Engineering	Course Type	Theory

Course Objectives:

1. To provide students with foundational knowledge of essential software testing techniques, including unit testing, boundary value testing, equivalence class testing, and decision table-based testing.
2. To introduce students to life cycle-based and agile testing methodologies to ensure software quality throughout various stages of development and learn about test management, planning, and the use of automated testing tools.
3. To equip students with the skills to apply system testing and object-oriented testing techniques to identify and resolve software defects.
4. To foster Test-Driven Development Practices: Encourage students to adopt test-driven development to produce reliable and maintainable software solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction- Software Testing Terminology- STLC-Software Testing Methodology- Verification and Validation Activities-Verification – Verification of Requirements- Verification of High level Design- Verification of Low level Design - How to verify Code- Validation, Functional testing, Non Functional Testing, SDLC and STLC relationship.	9
2	Testing Techniques: Dynamic Testing: Black Box Testing, Dynamic Testing: White Box Testing, Static Testing, Validation Activities, Regression Testing, Unit Testing, Integration Testing, Sanity Testing, Smoke Testing,	9

	Performance Testing, Load Testing, Stress Testing, Usability Testing, Compatibility Testing	
3	Managing The Test Process:Test Management, Software Metrics, Testing Metrics for Monitoring and Controlling the Testing Process, Efficient Test Suite Management, Testing Process Maturity Models, Introduction to Test Management Tool - Jira	**9**
4	Test Automation: Automation and Testing Tools - Testing For Specialized Environment: Testing Object Oriented Software, Testing Web based Systems, Tracking The Bug: Debugging	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
● 2 Questions from each module. ● Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	● Each question carries 9 marks. ● Two questions will be given from each module, out of which 1 question should be answered. ● Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain software testing life cycle (STLC) to ensure comprehensive software quality.	**K2**
CO2	Apply various software testing techniques to ensure comprehensive validation and reliability of software systems.	**K3**
CO3	Perform the software testing process effectively to monitor, control, and improve the testing process.	**K3**
CO4	Apply test automation strategies to effectively test object-oriented software and web-based systems	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	2	-	-	-	-	-	2	2
CO2	2	3	2	2	2	-	-	-	-	-	2	2
CO3	2	3	3	2	2	-	-	-	-	2	2	2
CO4	3	3	3	2	3	-	-	-	2	2	2	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Software testing : principles and practices	Naresh Chauhan	Oxford University Press	2nd edition, 2016
2	Software Testing: A Craftsman's Approach	Paul C. Jorgensen	CRC Press	4th Edition, 2014
3	Introduction to Software Testing,	Paul Ammann and Jeff Offutt	Cambridge University Press.	2008

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	The Art of Software Testing	Glenford J. Myers, Tom Badgett, Corey Sandler	John Wiley & Sons Publication	3rd Edition, 2012
2	Software Testing and Quality Assurance: Theory and Practice	KshirasagarNaik and PriyadarshiTripathy	Wiley-Spektrum	1st edition, 2008

Video Links (NPTEL, SWAYAM…)	
Sl. No	**Link ID**
1	https://www.youtube.com/playlist?list=PLrpK1inhO61VDiW_RBhkizmTYyUE0eoAF
2	https://nptel.ac.in/courses/106105150
3	https://www.youtube.com/watch?v=BlJEP7XG5iY
4	https://youtu.be/b1Qpvqm0UAo?si=M6ALob-iLJ9E0i_r

SEMESTER S8

ADHOC AND WIRELESS SENSOR NETWORKS

Course Code	**PEITT863**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	**ESE Marks**	60
Credits	3	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	PCITT402: Computer Networks	**Course Type**	Theory

Course Objectives:

1. To introduce the fundamental concepts and characteristics of Adhoc networks.
2. To understand the routing protocols and techniques used in Adhoc networks.
3. To explore the architecture and protocols of Wireless Sensor Networks (WSNs).
4. To examine the applications, challenges, and security concerns in Adhoc and Wireless Sensor Networks

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Adhoc Networks: Definition and Characteristics, Types of Adhoc Networks: MANETs, VANETs, and FANETs, Challenges in Adhoc Networks: Scalability, Mobility, Energy Efficiency, Medium Access Control in Adhoc Networks, Case Studies and Examples.	9
2	Introduction to Routing in Adhoc Networks, Proactive, Reactive, and Hybrid Routing Protocols, AODV, DSR, OLSR, and TORA Protocols, Quality of Service (QoS) in Adhoc Networks, Performance Analysis of Routing Protocols.	9
3	Introduction to Wireless Sensor Networks: Architecture and Protocols, Sensor Node Architecture and Design, Communication Protocols in WSNs: MAC, Routing, and Transport Layer Protocols, Energy Management in WSNs, WSN Applications in Environmental Monitoring, Healthcare, and Industrial Automation.	9
4	Key Applications of Adhoc Networks and WSNs, Challenges in Deployment and Maintenance, Security Issues in Adhoc Networks and WSNs: Attacks, Intrusion Detection, Privacy Concerns and Mitigation Techniques, Future Trends and Emerging Technologies in Adhoc and WSNs.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Apply the principles of Adhoc networks to design and implement basic network configurations.	**K3**
CO2	Analyze and implement routing protocols specific to Adhoc networks.	**K3**
CO3	Design and evaluate Wireless Sensor Networks for various applications	**K3**
CO4	Assess the challenges, applications, and security measures in Adhoc and Wireless Sensor Networks.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	-	-	-	-	-	-	-	3
CO2	3	3	3	3	-	-	-	-	-	-	-	3
CO3	3	3	3	3	-	-	-	-	-	-	-	3
CO4	3	3	2	2	-	-	-	-	-	-	-	3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Ad Hoc and Sensor Networks: Theory and Applications	Carlos Corderio, Dharma P. Agrawal	World Scientific	2nd Edition, 2011

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Module I: Mobile Ad Hoc Networking: The Cutting Edge Directions	Stefano Basagni, Marco Conti, Silvia Giordano, Ivan Stojmenovic.	Wiley-IEEE Press	2nd Edition, 2013
2	Module II: Ad Hoc Wireless Networks: Architectures and Protocols	C. Siva Ram Murthy, B. S. Manoj	Pearson Education	1st Edition, 2004
3	Module III: Wireless Sensor Networks: An Information Processing Approach	Feng Zhao, Leonidas J. Guibas	Morgan Kaufmann	1st Edition, 2004
4	Module IV: Security in Wireless Ad Hoc and Sensor Networks	Erdal Cayirci, Chunming Rong	Wiley	1st Edition, 2009

Video Links (NPTEL, SWAYAM…)	
Sl. No	**Link ID**
1	https://www.youtube.com/watch?v=Jh46Ivv6gQ0
2	https://www.youtube.com/watch?v=YFPln8wDMEg
3	https://www.youtube.com/watch?v=B3ZNjE_gO6w
4	https://www.youtube.com/watch?v=zQWE7HEBOck

SEMESTER S8
SEMANTIC WEB

Course Code	PEITT864	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBITT504 - Web Application Development	Course Type	Theory

Course Objectives:

1. To introduce the foundational concepts and architecture of the Semantic Web.
2. To explore RDF and SPARQL as the core technologies for representing and querying data on the Semantic Web.
3. To provide an understanding of Ontology design and the use of OWL for Semantic Web applications.
4. To investigate the practical applications of the Semantic Web and the role of reasoning techniques.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction to the Semantic Web and RDF:** Evolution of the Web and the need for the Semantic Web, Semantic Modeling, Semantic Web Technologies: RDF, RDFS, OWL, and SPARQL RDF (Resource Description Framework): Syntax , Semantics URI, Vocabularies, Metadata and the Semantic Web	9
2	**RDF and SPARQL:** Semantic Web Application Architecture, RDF and Interfacing, RDF Schema, RDFS-Plus, SKOS,FOAF, The Semantic Web, RDF, and Linked Data (and SPARQL) , SPARQL Queries- Copying, Creating, and Converting Data (and Finding Bad Data)	9
3	**Ontologies and OWL:** Introduction to Ontologies: Concepts and Importance, Ontology Design Principles Basic OWL(Web Ontology Language), Ontologies in OWL, OWL Formal Semantics, Ontologies and Rules, Ontology Engineering	9
4	**Semantic Web Applications:** Applying Machine Reasoning and Learning in Real World Applications, Case Studies of Semantic Web Applications - Achieving Balance Between Innovation and Security in the Cloud With Artificial Intelligence of Things: Semantic Web Control Models- Traffic: An Intelligent System for Detecting Traffic Events Based on Ontologies- Multi-Factor Authentication Web	9

	Security System Based on Facial Recognition, One Time Password, and Hashed Secure Question- Enhancing Usability and Control in Artificial Intelligence of Things Environments (AIoT) Through Semantic Web Control Models- Emerging Trends in Artificial Intelligence of Things With Machine Learning and Semantic Web Convergence	

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE):
In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)
At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the principles of the Semantic Web to describe its layered architecture.	**K3**
CO2	Construct and query RDF data using SPARQL in Semantic Web environments.	**K3**
CO3	Develop and implement ontologies using OWL for semantic data representation.	**K3**
CO4	Utilize reasoning techniques in Semantic Web applications to solve complex problems.	**K3**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	3	-	-	-	-	1	1	2
CO2	3	2	3	3	3	-	-	-	-	1	2	2
CO3	3	2	3	3	3	-	-	-	-	1	2	2
CO4	3	3	3	3	3	-	-	-	-	1	1	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Semantic Web for the Working Ontologist: Effective Modeling in RDFS and OWL	Dean Allemang, James Hendler	Morgan Kaufmann	2nd Edition, 2011
2	Foundations of Semantic Web Technologies	Pascal Hitzler, Markus Krötzsch, Sebastian Rudolph	Chapman and Hall/CRC	1st Edition, 2009

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Learning SPARQL	Bob DuCharme	O'Reilly Media	2nd Edition, 2013
2	Ontology Engineering in a Networked World	Mari Carmen Suárez-Figueroa, Asunción Gómez-Pérez, Enrico Motta	Springer	1st Edition, 2012
3	Reasoning Web: Logical Foundation of Knowledge Graph Construction and Query Answering	Gerhard Lakemeyer, Bernhard Nebel	Springer	1st Edition, 2015
4	Semantic Web Technologies and Applications in Artificial Intelligence of Things	Fernando Ortiz-Rodriguez, Amed Leyva-Mederos, Sanju Tiwari, Ania R. Hernandez-Quintana, Jose L. and Martinez-Rodriguez	IGI Global	May, 2024

Video Links (NPTEL, SWAYAM…)	
Sl. No	**Link ID**
1	https://www.youtube.com/watch?v=sLnbDyG2z3Q
2	https://youtu.be/8LBS5S3-4bI
3	https://youtu.be/HEQDRWMK06I
4	https://www.youtube.com/watch?v=3JZ7l_OeQyE

SEMESTER S8

ROBOTICS AND AUTOMATION

Course Code	PEITT865	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCITT501: Machine Learning, PCITT602: Advanced Artificial Intelligence	Course Type	Theory

Course Objectives:

1. To introduce students to the fundamental concepts and history of robotics, and familiarize them with basic robotic systems and programming.
2. To provide an in-depth understanding of the kinematics and dynamics of robotic systems, and to develop skills in trajectory planning and control.
3. To introduce automation concepts and industrial robots, focusing on their application in modern manufacturing and production systems.
4. To explore advanced topics in robotics and automation, including machine learning, human-robot interaction, and future trends.
5. To develop students' abilities to critically analyze, evaluate, and design complex robotics and automation systems by applying advanced concepts, tools, and frameworks.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**History and Overview of Robotics:** Evolution of robotics, types of robots, and applications. **Fundamental Concepts in Robotics:** Kinematics, dynamics, control systems, sensors, and actuators. **Robot Classification:** Based on geometry, degrees of freedom, and application areas. **Basic Programming for Robots:** Introduction to robot programming languages and simulators (e.g., Python, ROS basics).	9

2	**History and Overview of Robotics:** Evolution of robotics, types of robots, and applications. **Fundamental Concepts in Robotics:** Kinematics, dynamics, control systems, sensors, and actuators. **Robot Classification:** Based on geometry, degrees of freedom, and application areas. **Basic Programming for Robots:** Introduction to robot programming languages and simulators (e.g., Python, ROS basics).	9
3	**Introduction to Automation:** Concepts of automation, types of automation (fixed, programmable, flexible). **Industrial Robots:** Types of industrial robots, applications in manufacturing, assembly, and inspection. **PLC and SCADA Systems:** Basics of Programmable Logic Controllers (PLCs), Supervisory Control and Data Acquisition (SCADA) systems. **Sensors and Actuators in Automation:** Types of sensors (proximity, vision, force), actuators (electric, hydraulic, pneumatic).	9
4	**Machine Learning in Robotics:** Introduction to machine learning algorithms and their applications in robotics (e.g., vision, decision-making). **Human-Robot Interaction (HRI):** Fundamentals of HRI, collaborative robots (cobots), safety in HRI. **Robot Operating System (ROS):** Advanced ROS concepts, ROS packages for manipulation and navigation. **Future Trends in Robotics and Automation:** Autonomous systems, robotics in healthcare, AI-driven automation.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Examination (Written)	Evaluate Level Assessment	Analyze Level Assessment	Total
5	15	10	10	40

Evaluate and Analyze Level Assessment [20 Marks]

Students should evaluate and analyze a real-world optimization problem, assess the proposed solutions, provide a conclusion on which solution is most appropriate for the problem.

Criteria for evaluation:

1. **Case Study Analysis (K4 - 4 points)**

   Assignment:

   - *Provide students with a case study of a real-world robotics or automation system. This could involve a specific robot used in manufacturing, a robotic surgery system, or an automated warehouse system.*
   - ***Task**: Ask students to analyze the system by breaking it down into its components (e.g., sensors, actuators, control systems, software). They should evaluate how each component contributes to the overall functionality and how these components interact with each other.*
   - ***Expected Output:** A detailed report where students identify key elements, explain their roles, and analyze the strengths and weaknesses of the system's design.*

2. **Comparative Evaluation of Robotics Frameworks (K4 – 4 Points)**

   Project:

   - *Have students work on a project where they compare two different robotics frameworks or automation tools (e.g., ROS vs. a proprietary robotics framework, or different PLC brands in automation).*
   - ***Task:** They should evaluate the frameworks/tools based on criteria such as ease of use, scalability, cost, community support, and performance in specific tasks.*
   - ***Expected Output:** A comparative study report or presentation that not only compares the tools but also provides a well-reasoned conclusion about which tool is better suited for particular applications, supported by evidence and critical evaluation.*

3. **Design and Critique (K5 - Evaluating)**

   Design Task:

   - *Ask students to design a simple robotic system or automation process for a given problem, such as automating a repetitive task in a manufacturing line or designing a robot for a specific healthcare application.*

- ***Task:*** *Once the design is completed, have them critique their own design by identifying potential weaknesses, risks, and areas for improvement. This self-evaluation should consider factors like efficiency, safety, and cost-effectiveness.*
- ***Expected Output:*** *A design document with a critical evaluation section where students justify their design choices and propose enhancements based on their evaluation.*

4. **Simulation-Based Problem Solving (K5 - 4 points)**

   Lab Task:

- *Use a robotics simulation software (e.g., Gazebo with ROS) where students are given a scenario with specific goals (e.g., navigate an environment, complete a task).*
- ***Task:*** *Students must analyze the problem, develop a solution within the simulation environment, and then evaluate the effectiveness of their solution, including how they could optimize the process or improve performance.*
- ***Expected Output:*** *A simulation report or demonstration video, accompanied by an analysis of the performance and a critique of the solution, with suggestions for improvements.*

5. **Peer Review (K4- 2 points, K5 – 2 points)**

   Activity:

- *Organize a peer review session where students present their projects to their peers. Each student group reviews another group's project, analyzing its strengths and weaknesses.*
- ***Task:*** *Students must provide constructive feedback and evaluate the feasibility and effectiveness of the approach taken by their peers, based on criteria such as technical soundness, innovation, and potential for real-world application.*
- ***Expected Output:*** *Written feedback and evaluation reports from peers **(K4)**, followed by a discussion where students defend their design decisions based on the critiques received **(K5)**.*

Scoring:

1. **Accomplished (4 points)**: Exceptional analysis, clear implementation, and depth of understanding.
2. **Competent (3 points)**: Solid performance with minor areas for improvement.
3. **Developing (2 points)**: Adequate effort but lacks depth or clarity.
4. **Minimal (1 point)**: Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
● 2 Questions from each module. ● Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	● Each question carries 9 marks. ● Two questions will be given from each module, out of which 1 question should be answered. ● Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply basic principles of robotics to program and simulate simple robotic tasks.	**K3**
CO2	Apply kinematic and dynamic models to design and control robotic manipulators.	**K3**
CO3	Implement automation systems using industrial robots, PLCs, and SCADA systems.	**K3**
CO4	Apply advanced techniques in robotics and automation to develop innovative solutions in emerging areas.	**K3**
CO5	Critically evaluate and optimize robotics and automation systems by analyzing their design, performance, and impact.	**K5**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	-	2	-	-	-	2	-	-	-
CO2	3	3	3	2	2	-	2	-	3	-	2	-
CO3	3	2	3	3	2	2	3	-	3	2	3	-
CO4	3	3	3	3	3	2	3	2	3	3	3	2
CO5	3	3	3	3	3	3	3	3	3	3	3	3

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Introduction to Robotics: Mechanics and Control	John J. Craig	Pearson	4th Edition, 2017

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Robotics: Modelling, Planning and Control	Bruno Siciliano, Lorenzo Sciavicco, Luigi Villani, Giuseppe Oriolo	Springer	2nd Edition, 2010
2	Robot Modeling and Control	Mark W. Spong, Seth Hutchinson, M. Vidyasagar	Wiley	1st Edition, 2005
3	Automation, Production Systems, and Computer-Integrated Manufacturing	Dafydd Stuttard, Marcus Pinto	Wiley	2nd Edition, 2011
4	Probabilistic Robotics	Sebastian Thrun, Wolfram Burgard, Dieter Fox	MIT Press	1st Edition, 2005

Video Links (NPTEL, SWAYAM…)	
Sl. No	**Link ID**
1	https://www.youtube.com/watch?v=rYWJdZ5qg6M&list=PLbRMhDVUMngcdUbBySzyzcPiFTYWr4rV_
2	https://www.youtube.com/watch?v=gXX-4rrB4tw&list=PLQ3sZ7NCnFlEej8AWH_BfO9W7xlirvK6l
3	https://www.youtube.com/watch?v=uOtdWHMKhnw
4	https://www.youtube.com/watch?v=LyC9RAYE96M

SEMESTER S8

COMPUTER VISION

Course Code	OEITT831	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Linear Algebra, Probability	Course Type	Theory

Course Objectives:

1. To equip students of image processing techniques for computer vision
2. Enable students to analyze various geometric techniques in computer vision
3. To enable students to summarize various feature detectors and descriptors and feature-matching algorithms
4. To equip students to describe motion analysis and object recognition techniques
5. To enable students to explore the applications of computer vision

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	**Introduction and Goals of Computer Vision** **Review of image processing techniques**: Image filtering, Image enhancement, Image segmentation, thresholding, Image Transforms, Color Image Processing, Mathematical morphology and use in texture analysis,Binary shape analysis and connectedness	9
2	**Measuring Light and Radiometry**: Reflectance and BRDF(basics only), Monocular and binocular imaging system, Geometric Transformation,Orthographic & Perspective Projection, Camera model-Basic Pinhole Camera model **Stereo vision**: introduction; concept of disparity and its relationship with depth	9
3	**Feature Detection and Description**: Harris Corner Detection, Canny Edge Detection, Blob Detection, Line and curve detection, Hough transform **Feature Descriptors**: Histogram of Oriented Gradients, Scale Invariant Feature Transform(SIFT),Mosaics and Snakes, Feature Matching algorithms:	9

	Brute Force Matcher, Need for feature detection and description and limitations	
4	Shape from X - Shape from shading, Photometric stereo. Occluding contour detection. **Motion Analysis**-Regularization theory, Optical Flow: brightness constancy equation, Lucas- Kanade method, Principles of Structure from motion(Introduction). **Object recognition**: Hough transforms and other simple object recognition methods **Application**: Face detection, Face recognition and Eigen faces	**9**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	**15**	**10**	**10**	**40**

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Implement fundamental image processing techniques required for computer vision	**K3**
CO2	Describe various geometric techniques in computer vision	**K2**
CO3	Illustrate various feature descriptors and image matching algorithms used in computer vision	**K3**
CO4	Explain various applications of computer vision that use motion tracking, object detection and recognition techniques	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	3	2										
CO3	3	2	2									
CO4	3	2	2									

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Computer vision: A modern approach	Forsyth and Ponce	Prentice Hall of India	Second Edition,2015
2	Computer Vision: Algorithms & Applications	Richard Szeleski	Springer	Second Edition,2022
3	Robot Vision	B K P Horn	McGraw-Hill	1986

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the**	**Edition and**

			Publisher	Year
1	Multiple View Geometry in Computer Vision	Richard Hartley and Andrew Zisserman	Cambridge University Press	Second Edition,2004
2	Computer & Machine Vision	E R Davies	Academic Press	Fourth Edition,2012

Video Links (NPTEL, SWAYAM…)	
Sl. No	**Link ID**
1	Link to an NPTEL course on Computer Vision: https://archive.nptel.ac.in/courses/106/105/106105216/
2	Computer Vision Models: https://udlbook.github.io/cvbook/

SEMESTER S8

DEEP LEARNING

Course Code	OEITT832	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	OEITT721	Course Type	Theory

Course Objectives:

1. To provide students the understanding of deep learning principles and neural network architectures.
2. To enable students to explore generative models and transfer learning, the latest trends in deep learning.

SYLLABUS

Module No.	Syllabus Description	Contact Hours (36 Hrs)
1	**Introduction to Deep Learning:** Traditional Machine Learning to Deep Learning, Neural Networks- Feedforward Neural Networks, Activation Functions -Sigmoid, ReLU, Tanh, Softmax, Training Neural Networks- Forward and Backward Propagation Algorithm, Optimization Algorithms (Gradient Descent, Stochastic Gradient Descent, Adam).	**9**
2	**Convolutional Neural Networks (CNNs):** CNN Basics- Convolutional Layers, Pooling Layers, CNN Architectures- LeNet, AlexNet, VGG, ResNet, Applications of CNNs.	**8**
3	**Recurrent Neural Networks (RNNs) and Transformers:** RNN Basics- RNN Structure, Backpropagation Through Time (BPTT), Vanishing and Exploding Gradients, Advanced RNN Architectures- Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), Transformers and Attention Mechanisms- Attention Mechanism Basics, Transformer Architecture (BERT, GPT).	**11**
4	**Generative Models and Transfer Learning:** Generative Adversarial Networks (GANs)- GAN Basics, Applications of GANs, Autoencoders- Architecture, Transfer Learning and Fine-Tuning- Pre-trained Models, Fine-Tuning Strategies.	**8**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):
In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)
At the end of the course students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Build basic Neural Network models using appropriate activation functions and optimization techniques.	**K3**
CO2	Apply Convolutional Neural Networks and their architectures to various tasks.	**K3**
CO3	Develop Recurrent Neural Networks and Transformer models.	**K3**
CO4	Explain the principles of Generative Adversarial Networks and Autoencoders.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2	2	-	-	-	-	2	-	-	2
CO2	1	3	2	2	-	-	-	-	2	-	-	2
CO3	1	2	3	3	-	-	-	-	2	-	-	2
CO4	1	2	3	3	-	-	-	-	1	-	-	2

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Deep Learning	Ian Goodfellow, Yoshua Bengio, and Aaron Courville	MIT Press	1st Edition, 2016

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Neural Networks and Deep Learning: A Textbook	Charu C. Aggarwal	Springer	1st Edition, 2018
2	Deep Learning for Computer Vision	Rajalingappaa Shanmugamani	Packt Publishing	1st Edition, 2018
3	Natural Language Processing with PyTorch	Delip Rao and Brian McMahan	O'Reilly Media	1st Edition, 2019
4	Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play	David Foster	O'Reilly Media	1st Edition, 2020

Video Links (NPTEL, SWAYAM…)	
Sl. No	**Link ID**
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106184/

SEMESTER S8

WEB DESIGNING

Course Code	OEITT833	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Gain a comprehensive understanding of web development, from structuring and styling web pages with HTML and CSS to adding interactivity with JavaScript and jQuery, and understanding server interactions and web hosting.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Website creation roles, Gearing up for web design, Web Page Addresses-URL. HTML: HTML document structure, identifying text elements, adding an image, Changing the look with a style sheet, marking up Elements: Text, Paragraph, Heading, Horizontal Rule, Lists, Links, href attribute, Linking to pages on the web, Linking within your site, Targeting a new browser window, Mail links. Images: image formats, img element, adding SVG elements, Responsive image markup. Table, Forms, Embedded media: iframe, object, video and audio, canvas.	8
2	Concept of CSS, Creating Style Sheet, CSS Properties, CSS Styling (Background, Text Format, Controlling Fonts), Working with block elements and objects, Working with Lists and Tables, CSS Id and Class, Box Model (Introduction, Border properties, Padding Properties, Margin properties), Navigation Bar, CSS Color, Creating page Layout and Site Designs, Positioning: Floating and positioning: normal flow, floating, fancy text wrap with CSS shapes, positioning basics, relative positioning, Absolute positioning, fixed positioning; CSS Layout with Flexbox and Grid. Responsive Web Design: RWD, The responsive recipe, choosing	10

	breakpoints, designing responsively, Transition, Transforms, and Animations: CSS transitions, CSS transforms, Keyframe animations	
3	**Introduction to JavaScript:** JavaScript, Adding JavaScript to a page, the anatomy of a script, the browser object, Events, putting it all together, Meet the DOM, Polyfills, JavaScript libraries **jQuery:** jQuery, A basic jQuery example, jQuery use, finding elements, jQuery selection, Getting element content, Updating elements, Changing content, Inserting elements, Adding new content, Getting and setting attributes, Getting and setting CSS properties, Using .each(), events, The event object, Effects, Animating CSS properties, Using animation, traversing the DOM, Working with forms, JavaScript libraries, jQuery and Ajax.	**10**
4	Introduction to XML: syntax, element, attribute, HttpRequest SVG: Drawing with XML, Features of SVG as XML, SVG tools, SVG production tips, Responsive SVG. Web Servers: Introduction, HTTP Transactions, Multitier Application Architecture, Client-Side Scripting versus Server-Side Scripting, Accessing Web Servers. Introduction to Web Publishing or Hosting.	**8**

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks **(8x3 =24marks)**	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub-divisions. **(4x9 = 36 marks)**	**60**

Course Outcomes (COs)

At the end of the course, students should be able to:

	Course Outcome	Bloom's Knowledge Level (KL)
CO1	Summarise HTML tags.	**K2**
CO2	Apply CSS to add presentation style to web pages.	**K3**
CO3	Apply JavaScript to add functionality to web pages.	**K3**
CO4	Understand the basics of XML, SVG, and web servers.	**K2**

Note*: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	3	-	-	-	2	-	-	2
CO2	3	2	2	-	3	-	-	-	2	-	-	
CO3	3	2	3	-	3	-	-	-	2	-	-	
CO4	3	2	2	-	3	-	-	-	2	-	-	

Note: *1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation*

Text Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	Learning Web Design	Jennifer Niederst Robbins	O’Reilly	Fifth/2018
2	JavaScript and JQuery: Interactive Front–End Web Development	Jon Duckett	Wiley	First/2014
3	MASTERING HTML, CSS & Java Script Web Publishing	Laura Lemay Rafe Colburn Jennifer Kyrnin	BPB Publications	First/2016
4	Internet and World Wide Web How To Program	Paul J. Deitel, Harvey M. Deitel, Abbey Deitel	Pearson Education	Fifth/2012

Reference Books				
Sl. No	**Title of the Book**	**Name of the Author/s**	**Name of the Publisher**	**Edition and Year**
1	HTML and CSS: Design and Build Websites	Jon Duckett	Wiley	1st Edition 2011
2	CSS: The Definitive Guide	Eric A. Meyer	O'Reilly Media	4th Edition 2017
3	CSS Secrets: Better Solutions to Everyday Web Design Problems	Lea Verou	O'Reilly Media	1st Edition 2015
4	Bulma: A Modern CSS Framework	Michael L. Smith	Packt Publishing	1st Edition 2019
5	JavaScript: The Good Parts	Douglas Crockford	O'Reilly Media	1st Edition 2008
6	Eloquent JavaScript: A Modern Introduction to Programming	Marijn Haverbeke	No Starch Press	3rd Edition 2018

Video Links (NPTEL, SWAYAM…)	
Sl. No	**Link ID**
1	https://archive.nptel.ac.in/courses/106/106/106106156/